Part One: DistilRoBERTa and RoBERTa classification

1. Setting up the models

In [46]:
```
!pip install transformers
!pip install opendatasets
!pip install pandas
```

```
Requirement already satisfied: transformers in /usr/local/lib/python3.12/dist-packages (5.0.0)
Requirement already satisfied: filelock in /usr/local/lib/python3.12/dist-packages (from transformers) (3.29.0)
Requirement already satisfied: huggingface-hub<2.0,>=1.3.0 in /usr/local/lib/python3.12/dist-packages (from transformers) (1.11.0)
Requirement already satisfied: numpy>=1.17 in /usr/local/lib/python3.12/dist-packages (from transformers) (2.0.2)
Requirement already satisfied: packaging>=20.0 in /usr/local/lib/python3.12/dist-packages (from transformers) (26.1)
Requirement already satisfied: pyyaml>=5.1 in /usr/local/lib/python3.12/dist-packages (from transformers) (6.0.3)
Requirement already satisfied: regex!=2019.12.17 in /usr/local/lib/python3.12/dist-packages (from transformers) (2025.11.3)
Requirement already satisfied: tokenizers<=0.23.0,>=0.22.0 in /usr/local/lib/python3.12/dist-packages (from transformers) (0.22.2)
Requirement already satisfied: typer-slim in /usr/local/lib/python3.12/dist-packages (from transformers) (0.24.0)
Requirement already satisfied: safetensors>=0.4.3 in /usr/local/lib/python3.12/dist-packages (from transformers) (0.7.0)
Requirement already satisfied: tqdm>=4.27 in /usr/local/lib/python3.12/dist-packages (from transformers) (4.67.3)
Requirement already satisfied: fsspec>=2023.5.0 in /usr/local/lib/python3.12/dist-packages (from huggingface-hub<2.0,>=1.3.0->transformers) (2025.3.0)
Requirement already satisfied: hf-xet<2.0.0,>=1.4.3 in /usr/local/lib/python3.12/dist-packages (from huggingface-hub<2.0,>=1.3.0->transformers) (1.4.3)
Requirement already satisfied: httpx<1,>=0.23.0 in /usr/local/lib/python3.12/dist-packages (from huggingface-hub<2.0,>=1.3.0->transformers) (0.28.1)
Requirement already satisfied: typer in /usr/local/lib/python3.12/dist-packages (from huggingface-hub<2.0,>=1.3.0->transformers) (0.24.2)
Requirement already satisfied: typing-extensions>=4.1.0 in /usr/local/lib/python3.12/dist-packages (from huggingface-hub<2.0,>=1.3.0->transformers) (4.15.0)
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Requirement already satisfied: certifi in /usr/local/lib/python3.12/dist-packages (from httpx<1,>=0.23.0->huggingface-hub<2.0,>=1.3.0->transformers) (2026.4.22)
Requirement already satisfied: httpcore==1.* in /usr/local/lib/python3.12/dist-packages (from httpx<1,>=0.23.0->huggingface-hub<2.0,>=1.3.0->transformers) (1.0.9)
Requirement already satisfied: idna in /usr/local/lib/python3.12/dist-packages (from httpx<1,>=0.23.0->huggingface-hub<2.0,>=1.3.0->transformers) (3.13)
Requirement already satisfied: h11>=0.16 in /usr/local/lib/python3.12/dist-packages (from httpcore==1.*->httpx<1,>=0.23.0->huggingface-hub<2.0,>=1.3.0->transformers) (0.16.0)
Requirement already satisfied: click>=8.2.1 in /usr/local/lib/python3.12/dist-packages (from typer-slim->huggingface-hub<2.0,>=1.3.0->transformers) (8.3.3)
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Requirement already satisfied: rich>=12.3.0 in /usr/local/lib/python3.12/dist-packages (from typer->huggingface-hub<2.0,>=1.3.0->transformers) (13.9.4)
Requirement already satisfied: annotated-doc>=0.0.2 in /usr/local/lib/python3.12/dist-packages (from typer->huggingface-hub<2.0,>=1.3.0->transformers) (0.0.4)
Requirement already satisfied: markdown-it-py>=2.2.0 in /usr/local/lib/python3.12/dist-packages (from rich>=12.3.0->typer->huggingface-hub<2.0,>=1.3.0->transformers) (4.0.0)
Requirement already satisfied: pygments<3.0.0,>=2.13.0 in /usr/local/lib/python3.12/dist-packages (from rich>=12.3.0->typer->huggingface-hub<2.0,>=1.3.0->transformers) (2.20.0)
Requirement already satisfied: mdurl~=0.1 in /usr/local/lib/python3.12/dist-packages (from markdown-it-py>=2.2.0->rich>=12.3.0->typer->huggingface-hub<2.0,>=1.3.0->transformers) (0.1.2)
Requirement already satisfied: opendatasets in /usr/local/lib/python3.12/dist-packages (0.1.22)
Requirement already satisfied: tqdm in /usr/local/lib/python3.12/dist-packages (from opendatasets) (4.67.3)
Requirement already satisfied: kaggle in /usr/local/lib/python3.12/dist-packages (from opendatasets) (2.0.2)
Requirement already satisfied: click in /usr/local/lib/python3.12/dist-packages (from opendatasets) (8.3.3)
Requirement already satisfied: bleach in /usr/local/lib/python3.12/dist-packages (from kaggle->opendatasets) (6.3.0)
Requirement already satisfied: jupytext in /usr/local/lib/python3.12/dist-packages (from kaggle->opendatasets) (1.19.1)
Requirement already satisfied: kagglesdk<1.0,>=0.1.20 in /usr/local/lib/python3.12/dist-packages (from kaggle->opendatasets) (0.1.20)
Requirement already satisfied: packaging in /usr/local/lib/python3.12/dist-packages (from kaggle->opendatasets) (26.1)
Requirement already satisfied: protobuf in /usr/local/lib/python3.12/dist-packages (from kaggle->opendatasets) (5.29.6)
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Requirement already satisfied: python-slugify in /usr/local/lib/python3.12/dist-packages (from kaggle->opendatasets) (8.0.4)
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Requirement already satisfied: nbformat in /usr/local/lib/python3.12/dist-packages (from jupytext->kaggle->opendatasets) (5.10.4)
Requirement already satisfied: pyyaml in /usr/local/lib/python3.12/dist-packages (from jupytext->kaggle->opendatasets) (6.0.3)
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Requirement already satisfied: charset_normalizer<4,>=2 in /usr/local/lib/python3.12/dist-packages (from requests->kaggle->opendatasets) (3.4.7)
Requirement already satisfied: idna<4,>=2.5 in /usr/local/lib/python3.12/dist-packages (from requests->kaggle->opendatasets) (3.13)
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Requirement already satisfied: jupyter-core!=5.0.*,>=4.12 in /usr/local/lib/python3.12/dist-packages (from nbformat->jupytext->kaggle->opendatasets) (5.9.1)
Requirement already satisfied: traitlets>=5.1 in /usr/local/lib/python3.12/dist-packages (from nbformat->jupytext->kaggle->opendatasets) (5.7.1)
Requirement already satisfied: attrs>=22.2.0 in /usr/local/lib/python3.12/dist-packages (from jsonschema>=2.6->nbformat->jupytext->kaggle->opendatasets) (26.1.0)
Requirement already satisfied: jsonschema-specifications>=2023.03.6 in /usr/local/lib/python3.12/dist-packages (from jsonschema>=2.6->nbformat->jupytext->kaggle->opendatasets) (2025.9.1)
Requirement already satisfied: referencing>=0.28.4 in /usr/local/lib/python3.12/dist-packages (from jsonschema>=2.6->nbformat->jupytext->kaggle->opendatasets) (0.37.0)
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Requirement already satisfied: platformdirs>=2.5 in /usr/local/lib/python3.12/dist-packages (from jupyter-core!=5.0.*,>=4.12->nbformat->jupytext->kaggle->opendatasets) (4.9.6)
Requirement already satisfied: typing-extensions>=4.4.0 in /usr/local/lib/python3.12/dist-packages (from referencing>=0.28.4->jsonschema>=2.6->nbformat->jupytext->kaggle->opendatasets) (4.15.0)
Requirement already satisfied: pandas in /usr/local/lib/python3.12/dist-packages (2.2.2)
Requirement already satisfied: numpy>=1.26.0 in /usr/local/lib/python3.12/dist-packages (from pandas) (2.0.2)
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Requirement already satisfied: tzdata>=2022.7 in /usr/local/lib/python3.12/dist-packages (from pandas) (2026.1)
Requirement already satisfied: six>=1.5 in /usr/local/lib/python3.12/dist-packages (from python-dateutil>=2.8.2->pandas) (1.17.0)
```

In [47]:
```
import os
os.environ["CUDA_LAUNCH_BLOCKING"] = "1"

import torch
import numpy as np

from transformers import pipeline

DistilRoBERTa_classifier = pipeline(
    "text-classification",
    model="j-hartmann/emotion-english-distilroberta-base",
    tokenizer="j-hartmann/emotion-english-distilroberta-base",
    device=-1
)


RoBERTa_classifier = pipeline(
    task="text-classification",
    model="SamLowe/roberta-base-go_emotions",
    tokenizer="SamLowe/roberta-base-go_emotions",
    device=-1,
    top_k=None)
```

```
Loading weights:   0%|          | 0/105 [00:00<?, ?it/s]
RobertaForSequenceClassification LOAD REPORT from: j-hartmann/emotion-english-distilroberta-base
Key                              | Status     |  |
---------------------------------+------------+--+-
roberta.embeddings.position_ids | UNEXPECTED |  |

Notes:
- UNEXPECTED    :can be ignored when loading from different task/architecture; not ok if you expect identical arch.
Loading weights:   0%|          | 0/201 [00:00<?, ?it/s]
```

```
RobertaForSequenceClassification LOAD REPORT from: SamLowe/roberta-base-go_emotions
Key                             | Status     |  |
--------------------------------+------------+--+-
roberta.embeddings.position_ids | UNEXPECTED |  |

Notes:
- UNEXPECTED    :can be ignored when loading from different task/architecture; not ok if you expect identical arch.
```

2. Importing the dataset

In [48]:
```python
import opendatasets as od
import pandas

od.download(
    "https://www.kaggle.com/datasets/mexwell/poem-dataset")
```

```
Skipping, found downloaded files in "./poem-dataset" (use force=True to force download)
```

In [49]:
```python
import pandas as pds

file =('/content/poem-dataset/final_df_emotions(remove-bias).csv')
newData = pds.read_csv(file)

newData.head()
```

Out[49]:

	poem content	pred	label	score	anger	disgust	fear	joy	neutral	sadness	surprise	age	type
0	Let the bird of loudest lay\r\nOn the sole Ara...	5	sadness	0.650738	0.133402	0.076618	0.094706	0.006762	0.032960	0.650738	0.004814	Renaissance	Mythology & Folklore
1	Sir Charles into my chamber coming in,\r\nWhen...	0	anger	0.264296	0.264296	0.145707	0.057977	0.262715	0.019884	0.244457	0.004963	Renaissance	Mythology & Folklore
2	Our vice runs beyond all that old men saw,\r\n...	0	anger	0.758054	0.758054	0.210444	0.003690	0.001755	0.013997	0.011483	0.000578	Renaissance	Mythology & Folklore
3	Lo I the man, whose Muse whilome did maske,\r\...	0	anger	0.896280	0.896280	0.009259	0.004949	0.002320	0.007623	0.078088	0.001481	Renaissance	Mythology & Folklore
4	Long have I longd to see my love againe,\r\nSt...	5	sadness	0.600543	0.027077	0.036344	0.075487	0.017212	0.187106	0.600543	0.056230	Renaissance	Mythology & Folklore

In [50]:
```python
poem_content = newData['poem content']

display(poem_content.head())
```

	poem content
0	Let the bird of loudest lay\r\nOn the sole Ara...
1	Sir Charles into my chamber coming in,\r\nWhen...
2	Our vice runs beyond all that old men saw,\r\n...
3	Lo I the man, whose Muse whilome did maske,\r\...
4	Long have I longd to see my love againe,\r\nSt...

dtype: object

In [51]:
```python
first_poem = poem_content.iloc[100]
first_label = newData['label'].iloc[100]

display(first_poem)
display(first_label)
```

```
'Forget not yet the tried intent\r\nOf such a truth as I have meant;\r\nMy great travail so gladly spent,\r\n            Forget not yet.\r\n\r\n   Forget not yet when first began\r\nThe weary life ye know, since whan\r\nThe suit, the service, none tell can;\r\n            Forget not yet.\r\n\r\n   Forget not yet the great assays,\r\nThe cruel wrong,
the scornful ways;\r\nThe painful patience in denays,\r\n            Forget not yet.\r\n\r\n   Forget not yet, forget not this,\r\nHow long ago hath been and is\r\nThe mind that never meant amiss;\r\n            Forget not yet.\r\n\r\n   Forget not then thine own approved,\r\nThe which so long hath thee so loved,\r\nWhose steadfast faith yet never mo
ved;\r\n            Forget not this.'
'sadness'
```

3. Making predictions

In [52]:
```python
import numpy as np

true = newData['label']

samples = newData['poem content']
samples = [s.replace('\r\n', '\n').strip() for s in samples]



DistilRoBERTa_pred_labels = []
for s in samples:
    result = DistilRoBERTa_classifier(
        s,
        truncation=True,
        padding=True
    )
    DistilRoBERTa_pred_labels.append(result[0]['label'])
```

In [53]:
```python
RoBERTa_pred_labels = []

RoBERTa_classifier(first_poem)


for s in samples:
    result = RoBERTa_classifier(
        s,
        truncation=True,
        padding=True
    )
    RoBERTa_pred_labels.append(result[0][0]['label'])
```

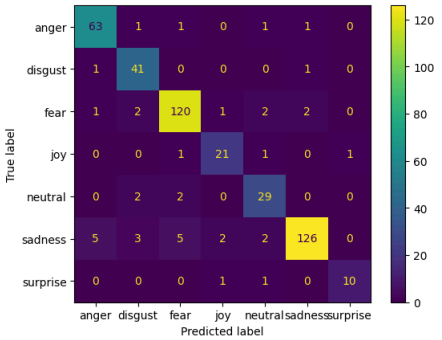
4. Mapping and Metrics

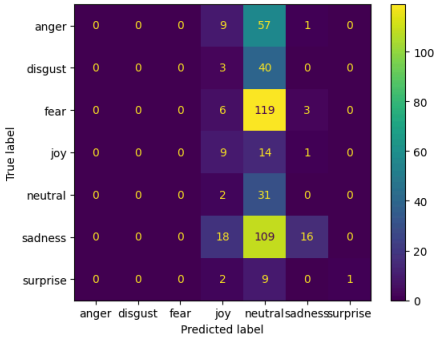
In [54]:
```python
from sklearn import metrics


label_map = {
    'admiration': 'joy',
    'amusement': 'joy',
    'anger': 'anger',
    'annoyance': 'anger',
    'approval': 'joy',
    'caring': 'joy',
    'confusion': 'neutral',
```

```
    'curiosity': 'joy',
    'desire': 'joy',
    'disappointment': 'sadness',
    'disapproval': 'disgust',
    'disgust': 'disgust',
    'embarrassment': 'fear',
    'excitement': 'joy',
    'fear': 'fear',
    'gratitude': 'joy',
    'grief': 'sadness',
    'joy': 'joy',
    'love': 'joy',
    'nervousness': 'fear',
    'optimism': 'joy',
    'pride': 'joy',
    'realization': 'surprise',
    'relief': 'joy',
    'remorse': 'sadness',
    'sadness': 'sadness',
    'surprise': 'surprise',
    'neutral': 'neutral',
}

RoBERTa_pred_labels = [label_map[p] for p in RoBERTa_pred_labels]
poem_labels = ["anger", "disgust", "fear", "joy", "neutral", "sadness", "surprise"]

def pred_metrics(true, pred):
    return {
        "f1": metrics.f1_score(true, pred, zero_division=0, average="macro"),
        "accuracy": metrics.accuracy_score(true, pred),
        "precision": metrics.precision_score(true, pred, zero_division=0, average="macro"),
        "recall": metrics.recall_score(true, pred, zero_division=0, average="macro"),
        "weighted f1": metrics.f1_score(true, pred, zero_division=0, average="weighted"),
    }

print(pred_metrics(true, DistilRoBERTa_pred_labels))
print(pred_metrics(true, RoBERTa_pred_labels))

DistilRoBERTa_labels = [DistilRoBERTa_classifier.model.config.id2label[i]
          for i in range(len(DistilRoBERTa_classifier.model.config.id2label))]

DistilRoBERTa_cm = metrics.confusion_matrix(true, DistilRoBERTa_pred_labels, labels=DistilRoBERTa_labels)
RoBERTa_cm = metrics.confusion_matrix(true, RoBERTa_pred_labels, labels=poem_labels)

DistilRoBERTa_disp = metrics.ConfusionMatrixDisplay(DistilRoBERTa_cm, display_labels=DistilRoBERTa_labels)
DistilRoBERTa_disp.plot()

RoBERTa_disp = metrics.ConfusionMatrixDisplay(RoBERTa_cm, display_labels=poem_labels)
RoBERTa_disp.plot()
```

```
{'f1': 0.8907470321224287, 'accuracy': 0.9111111111111111, 'precision': 0.88440635512776, 'recall': 0.8999324246851146, 'weighted f1': 0.9116109787946534}
{'f1': 0.10657555491780334, 'accuracy': 0.12666666666666668, 'precision': 0.28962463236331093, 'recall': 0.21565934065934064, 'weighted f1': 0.09029426779377311}
```

Out[54]: <sklearn.metrics._plot.confusion_matrix.ConfusionMatrixDisplay at 0x782d1ec0f650>





Part Two: Fine tuning DistilBERT and RoBERTa using dataset

1. Data Preparation for Fine-Tuning

We'll use the `datasets` library to convert our DataFrame and `LabelEncoder` to handle the target classes.

In [25]:
```
!pip install datasets

from datasets import Dataset
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import LabelEncoder
```

```python
# Map our text labels to integers
le = LabelEncoder()
newData['label_num'] = le.fit_transform(newData['label'])
label_list = le.classes_.tolist()

# Split into training and validation sets
train_df, val_df = train_test_split(newData, test_size=0.15, random_state=42)

train_dataset = Dataset.from_pandas(train_df[['poem content', 'label_num']])
val_dataset = Dataset.from_pandas(val_df[['poem content', 'label_num']])
```

```
Requirement already satisfied: datasets in /usr/local/lib/python3.12/dist-packages (4.0.0)
Requirement already satisfied: filelock in /usr/local/lib/python3.12/dist-packages (from datasets) (3.29.0)
Requirement already satisfied: numpy>=1.17 in /usr/local/lib/python3.12/dist-packages (from datasets) (2.0.2)
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Requirement already satisfied: dill<0.3.9,>=0.3.0 in /usr/local/lib/python3.12/dist-packages (from datasets) (0.3.8)
Requirement already satisfied: pandas in /usr/local/lib/python3.12/dist-packages (from datasets) (2.2.2)
Requirement already satisfied: requests>=2.32.2 in /usr/local/lib/python3.12/dist-packages (from datasets) (2.32.4)
Requirement already satisfied: tqdm>=4.66.3 in /usr/local/lib/python3.12/dist-packages (from datasets) (4.67.3)
Requirement already satisfied: xxhash in /usr/local/lib/python3.12/dist-packages (from datasets) (3.6.0)
Requirement already satisfied: multiprocess<0.70.17 in /usr/local/lib/python3.12/dist-packages (from datasets) (0.70.16)
Requirement already satisfied: fsspec<=2025.3.0,>=2023.1.0 in /usr/local/lib/python3.12/dist-packages (from fsspec[http]<=2025.3.0,>=2023.1.0->datasets) (2025.3.0)
Requirement already satisfied: huggingface-hub>=0.24.0 in /usr/local/lib/python3.12/dist-packages (from datasets) (1.11.0)
Requirement already satisfied: packaging in /usr/local/lib/python3.12/dist-packages (from datasets) (26.1)
Requirement already satisfied: pyyaml>=5.1 in /usr/local/lib/python3.12/dist-packages (from datasets) (6.0.3)
Requirement already satisfied: aiohttp!=4.0.0a0,!=4.0.0a1 in /usr/local/lib/python3.12/dist-packages (from fsspec[http]<=2025.3.0,>=2023.1.0->datasets) (3.13.5)
Requirement already satisfied: hf-xet<2.0.0,>=1.4.3 in /usr/local/lib/python3.12/dist-packages (from huggingface-hub>=0.24.0->datasets) (1.4.3)
Requirement already satisfied: httpx<1,>=0.23.0 in /usr/local/lib/python3.12/dist-packages (from huggingface-hub>=0.24.0->datasets) (0.28.1)
Requirement already satisfied: typer in /usr/local/lib/python3.12/dist-packages (from huggingface-hub>=0.24.0->datasets) (0.24.2)
Requirement already satisfied: typing-extensions>=4.1.0 in /usr/local/lib/python3.12/dist-packages (from huggingface-hub>=0.24.0->datasets) (4.15.0)
Requirement already satisfied: charset_normalizer<4,>=2 in /usr/local/lib/python3.12/dist-packages (from requests>=2.32.2->datasets) (3.4.7)
Requirement already satisfied: idna<4,>=2.5 in /usr/local/lib/python3.12/dist-packages (from requests>=2.32.2->datasets) (3.13)
Requirement already satisfied: urllib3<3,>=1.21.1 in /usr/local/lib/python3.12/dist-packages (from requests>=2.32.2->datasets) (2.5.0)
Requirement already satisfied: certifi>=2017.4.17 in /usr/local/lib/python3.12/dist-packages (from requests>=2.32.2->datasets) (2026.4.22)
Requirement already satisfied: python-dateutil>=2.8.2 in /usr/local/lib/python3.12/dist-packages (from pandas->datasets) (2.9.0.post0)
Requirement already satisfied: pytz>=2020.1 in /usr/local/lib/python3.12/dist-packages (from pandas->datasets) (2025.2)
Requirement already satisfied: tzdata>=2022.7 in /usr/local/lib/python3.12/dist-packages (from pandas->datasets) (2026.1)
Requirement already satisfied: aiohappyeyeballs>=2.5.0 in /usr/local/lib/python3.12/dist-packages (from aiohttp!=4.0.0a0,!=4.0.0a1->fsspec[http]<=2025.3.0,>=2023.1.0->datasets) (2.6.1)
Requirement already satisfied: aiosignal>=1.4.0 in /usr/local/lib/python3.12/dist-packages (from aiohttp!=4.0.0a0,!=4.0.0a1->fsspec[http]<=2025.3.0,>=2023.1.0->datasets) (1.4.0)
Requirement already satisfied: attrs>=17.3.0 in /usr/local/lib/python3.12/dist-packages (from aiohttp!=4.0.0a0,!=4.0.0a1->fsspec[http]<=2025.3.0,>=2023.1.0->datasets) (26.1.0)
Requirement already satisfied: frozenlist>=1.1.1 in /usr/local/lib/python3.12/dist-packages (from aiohttp!=4.0.0a0,!=4.0.0a1->fsspec[http]<=2025.3.0,>=2023.1.0->datasets) (1.8.0)
Requirement already satisfied: multidict<7.0,>=4.5 in /usr/local/lib/python3.12/dist-packages (from aiohttp!=4.0.0a0,!=4.0.0a1->fsspec[http]<=2025.3.0,>=2023.1.0->datasets) (6.7.1)
Requirement already satisfied: propcache>=0.2.0 in /usr/local/lib/python3.12/dist-packages (from aiohttp!=4.0.0a0,!=4.0.0a1->fsspec[http]<=2025.3.0,>=2023.1.0->datasets) (0.4.1)
Requirement already satisfied: yarl<2.0,>=1.17.0 in /usr/local/lib/python3.12/dist-packages (from aiohttp!=4.0.0a0,!=4.0.0a1->fsspec[http]<=2025.3.0,>=2023.1.0->datasets) (1.23.0)
Requirement already satisfied: anyio in /usr/local/lib/python3.12/dist-packages (from httpx<1,>=0.23.0->huggingface-hub>=0.24.0->datasets) (4.13.0)
Requirement already satisfied: httpcore==1.* in /usr/local/lib/python3.12/dist-packages (from httpx<1,>=0.23.0->huggingface-hub>=0.24.0->datasets) (1.0.9)
Requirement already satisfied: h11>=0.16 in /usr/local/lib/python3.12/dist-packages (from httpcore==1.*->httpx<1,>=0.23.0->huggingface-hub>=0.24.0->datasets) (0.16.0)
Requirement already satisfied: six>=1.5 in /usr/local/lib/python3.12/dist-packages (from python-dateutil>=2.8.2->pandas->datasets) (1.17.0)
Requirement already satisfied: click>=8.2.1 in /usr/local/lib/python3.12/dist-packages (from typer->huggingface-hub>=0.24.0->datasets) (8.3.3)
Requirement already satisfied: shellingham>=1.3.0 in /usr/local/lib/python3.12/dist-packages (from typer->huggingface-hub>=0.24.0->datasets) (1.5.4)
Requirement already satisfied: rich>=12.3.0 in /usr/local/lib/python3.12/dist-packages (from typer->huggingface-hub>=0.24.0->datasets) (13.9.4)
Requirement already satisfied: annotated-doc>=0.0.2 in /usr/local/lib/python3.12/dist-packages (from typer->huggingface-hub>=0.24.0->datasets) (0.0.4)
Requirement already satisfied: markdown-it-py>=2.2.0 in /usr/local/lib/python3.12/dist-packages (from rich>=12.3.0->typer->huggingface-hub>=0.24.0->datasets) (4.0.0)
Requirement already satisfied: pygments<3.0.0,>=2.13.0 in /usr/local/lib/python3.12/dist-packages (from rich>=12.3.0->typer->huggingface-hub>=0.24.0->datasets) (2.20.0)
Requirement already satisfied: mdurl~=0.1 in /usr/local/lib/python3.12/dist-packages (from markdown-it-py>=2.2.0->rich>=12.3.0->typer->huggingface-hub>=0.24.0->datasets) (0.1.2)
```

2. Tokenization and Model Setup

We will initialize the base models and tokenize the poem content.

In [26]:

```python
from transformers import AutoTokenizer, AutoModelForSequenceClassification, TrainingArguments, Trainer

def get_tokenized_dataset(model_name, dataset):
    tokenizer = AutoTokenizer.from_pretrained(model_name)
    def tokenize_func(examples):
        result = tokenizer(examples['poem content'], padding='max_length', truncation=True)
        result['labels'] = examples['label_num']
        return result
    return dataset.map(tokenize_func, batched=True), tokenizer

# Setup for DistilBERT
db_model_name = "distilbert-base-uncased"
train_db, db_tokenizer = get_tokenized_dataset(db_model_name, train_dataset)
val_db, _ = get_tokenized_dataset(db_model_name, val_dataset)

# Setup for RoBERTa
rb_model_name = "roberta-base"
train_rb, rb_tokenizer = get_tokenized_dataset(rb_model_name, train_dataset)
val_rb, _ = get_tokenized_dataset(rb_model_name, val_dataset)
```

```
config.json:   0%|          | 0.00/483 [00:00<?, ?B/s]
tokenizer_config.json:   0%|          | 0.00/48.0 [00:00<?, ?B/s]
vocab.txt: 0.00B [00:00, ?B/s]
tokenizer.json: 0.00B [00:00, ?B/s]
Map:   0%|          | 0/382 [00:00<?, ? examples/s]
Map:   0%|          | 0/68 [00:00<?, ? examples/s]
config.json:   0%|          | 0.00/481 [00:00<?, ?B/s]
tokenizer_config.json:   0%|          | 0.00/25.0 [00:00<?, ?B/s]
vocab.json: 0.00B [00:00, ?B/s]
merges.txt: 0.00B [00:00, ?B/s]
tokenizer.json: 0.00B [00:00, ?B/s]
Map:   0%|          | 0/382 [00:00<?, ? examples/s]
Map:   0%|          | 0/68 [00:00<?, ? examples/s]
```

3. Fine-Tuning Execution

Now we define the Trainer and start the training process for DistilBERT. (We can repeat this for RoBERTa next).

In [27]:

```python
def compute_metrics(eval_pred):
    logits, labels = eval_pred
    predictions = np.argmax(logits, axis=-1)
    return pred_metrics(le.inverse_transform(labels), le.inverse_transform(predictions))

training_args = TrainingArguments(
    output_dir="./results",
    eval_strategy="epoch",
    learning_rate=2e-5,
    per_device_train_batch_size=16,
    num_train_epochs=10,
    weight_decay=0.05,
)

# Initialize DistilBERT for Fine-Tuning
ft_distilbert = AutoModelForSequenceClassification.from_pretrained(db_model_name, num_labels=len(label_list))

trainer_db = Trainer(
    model=ft_distilbert,
```

```
    args=training_args,
    train_dataset=train_db,
    eval_dataset=val_db,
    compute_metrics=compute_metrics,
)

print("Starting Fine-tuning for DistilBERT...")
trainer_db.train()
```

```
model.safetensors:   0%|          | 0.00/268M [00:00<?, ?B/s]
Loading weights:   0%|          | 0/100 [00:00<?, ?it/s]
DistilBertForSequenceClassification LOAD REPORT from: distilbert-base-uncased
Key                     | Status     |
------------------------+------------+-
vocab_layer_norm.bias   | UNEXPECTED |
vocab_projector.bias    | UNEXPECTED |
vocab_layer_norm.weight | UNEXPECTED |
vocab_transform.bias    | UNEXPECTED |
vocab_transform.weight  | UNEXPECTED |
classifier.weight       | MISSING    |
pre_classifier.weight   | MISSING    |
pre_classifier.bias     | MISSING    |
classifier.bias         | MISSING    |

Notes:
- UNEXPECTED    :can be ignored when loading from different task/architecture; not ok if you expect identical arch.
- MISSING       :those params were newly initialized because missing from the checkpoint. Consider training on your downstream task.
Starting Fine-tuning for DistilBERT...
```

[240/240 01:20, Epoch 10/10]

Epoch	Training Loss	Validation Loss	F1	Accuracy	Precision	Recall	Weighted f1
1	No log	1.689141	0.065681	0.294118	0.042644	0.142857	0.135227
2	No log	1.615030	0.134526	0.382353	0.109869	0.185714	0.276965
3	No log	1.557292	0.158785	0.441176	0.126160	0.214286	0.326909
4	No log	1.529561	0.158632	0.441176	0.126292	0.214286	0.326595
5	No log	1.516971	0.163744	0.455882	0.130576	0.221429	0.337120
6	No log	1.526253	0.145292	0.411765	0.117965	0.200000	0.299131
7	No log	1.533857	0.145566	0.411765	0.119255	0.200000	0.299695
8	No log	1.523506	0.177482	0.426471	0.262690	0.215873	0.333051
9	No log	1.522022	0.165796	0.397059	0.254422	0.201587	0.308992
10	No log	1.509622	0.183058	0.441176	0.266538	0.223016	0.344530

```
Writing model shards:   0%|          | 0/1 [00:00<?, ?it/s]
```

Out[27]:
```
TrainOutput(global_step=240, training_loss=1.3189554850260417, metrics={'train_runtime': 82.1099, 'train_samples_per_second': 46.523, 'train_steps_per_second': 2.923, 'total_flos': 506070584094720.0, 'train_loss': 1.3189554850260417, 'epoch': 10.0})
```

4. Fine-Tuning RoBERTa

We will now follow the same process to fine-tune the `roberta-base` model.

In [28]:
```
ft_roberta = AutoModelForSequenceClassification.from_pretrained(rb_model_name, num_labels=len(label_list))

trainer_rb = Trainer(
    model=ft_roberta,
    args=training_args,
    train_dataset=train_rb,
    eval_dataset=val_rb,
    compute_metrics=compute_metrics,
)

print("Starting Fine-tuning for RoBERTa...")
trainer_rb.train()
```

```
model.safetensors:   0%|          | 0.00/499M [00:00<?, ?B/s]
Loading weights:   0%|          | 0/197 [00:00<?, ?it/s]
RobertaForSequenceClassification LOAD REPORT from: roberta-base
Key                             | Status     |
--------------------------------+------------+-
lm_head.layer_norm.weight       | UNEXPECTED |
roberta.embeddings.position_ids | UNEXPECTED |
lm_head.bias                    | UNEXPECTED |
lm_head.layer_norm.bias         | UNEXPECTED |
lm_head.dense.weight            | UNEXPECTED |
lm_head.dense.bias              | UNEXPECTED |
classifier.dense.bias           | MISSING    |
classifier.out_proj.bias        | MISSING    |
classifier.dense.weight         | MISSING    |
classifier.out_proj.weight      | MISSING    |

Notes:
- UNEXPECTED    :can be ignored when loading from different task/architecture; not ok if you expect identical arch.
- MISSING       :those params were newly initialized because missing from the checkpoint. Consider training on your downstream task.
Starting Fine-tuning for RoBERTa...
```

[240/240 02:48, Epoch 10/10]

Epoch	Training Loss	Validation Loss	F1	Accuracy	Precision	Recall	Weighted f1
1	No log	1.675425	0.065681	0.294118	0.042644	0.142857	0.135227
2	No log	1.588354	0.103203	0.323529	0.090709	0.157143	0.212478
3	No log	1.506433	0.148182	0.411765	0.120680	0.200000	0.305081
4	No log	1.396491	0.205574	0.470588	0.226399	0.242857	0.377992
5	No log	1.310773	0.309979	0.485294	0.409107	0.306349	0.434761
6	No log	1.329583	0.257136	0.470588	0.300144	0.284921	0.405932
7	No log	1.297184	0.320810	0.485294	0.309524	0.376190	0.429399
8	No log	1.230699	0.335426	0.485294	0.437758	0.342063	0.456520
9	No log	1.223616	0.364856	0.529412	0.481927	0.363492	0.489674
10	No log	1.215023	0.356798	0.514706	0.470776	0.356349	0.478981

```
Writing model shards:   0%|          | 0/1 [00:00<?, ?it/s]
```

Out[28]:
```
TrainOutput(global_step=240, training_loss=1.1035424550374349, metrics={'train_runtime': 168.8596, 'train_samples_per_second': 22.622, 'train_steps_per_second': 1.421, 'total_flos': 1005129352704000.0, 'train_loss': 1.1035424550374349, 'epoch': 10.0})
```

5. Final Evaluation

Once training is complete, we should evaluate all four models on the validation set to compare results.

In [29]:
```
print("DistilBERT Fine-tuned Metrics:")
display(trainer_db.evaluate())

print("\nRoBERTa Fine-tuned Metrics:")
display(trainer_rb.evaluate())
```

```
DistilBERT Fine-tuned Metrics:
```

[9/9 00:00]

```
{'eval_loss': 1.5096218585968018,
 'eval_f1': 0.18305764411027567,
 'eval_accuracy': 0.4411764705882353,
 'eval_precision': 0.2665379665379665,
 'eval_recall': 0.2230158730158730 3,
 'eval_weighted f1': 0.34453044375645,
 'eval_runtime': 0.5118,
 'eval_samples_per_second': 132.872,
 'eval_steps_per_second': 17.586,
 'epoch': 10.0}
RoBERTa Fine-tuned Metrics:
```

[9/9 00:00]

```
{'eval_loss': 1.2150230407714844,
 'eval_f1': 0.35679779158040026,
 'eval_accuracy': 0.5147058823529411,
 'eval_precision': 0.4707758907758908,
 'eval_recall': 0.35634920634920636,
 'eval_weighted f1': 0.4789814476515244,
 'eval_runtime': 0.966,
 'eval_samples_per_second': 70.395,
 'eval_steps_per_second': 9.317,
 'epoch': 10.0}
```

6. Summary Comparison of All Models

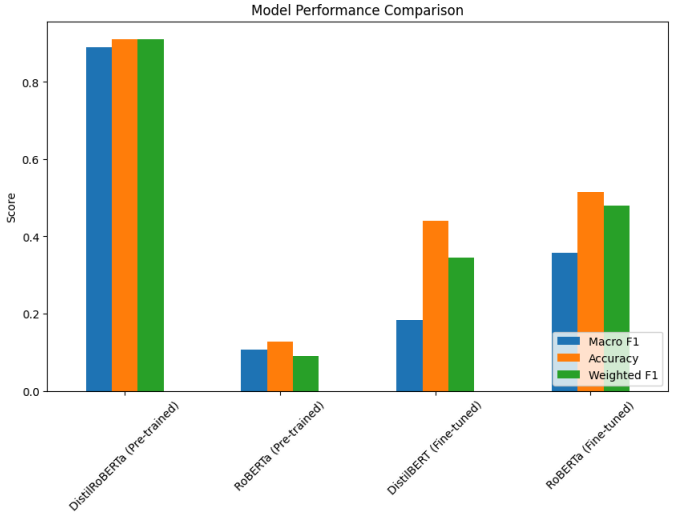
We will now collect the metrics from all four models to see how the pre-trained zero-shot models compare against the fine-tuned ones.

In [30]:
```
import pandas as pd
import matplotlib.pyplot as plt

# 1. Collect Pre-trained Metrics (already calculated earlier in notebook)
pre_trained_drb = pred_metrics(true, DistilRoBERTa_pred_labels)
pre_trained_rb = pred_metrics(true, RoBERTa_pred_labels)

# 2. Collect Fine-tuned Metrics
ft_db_results = trainer_db.evaluate()
ft_rb_results = trainer_rb.evaluate()

# Simplify keys for comparison
data = {
    "DistilRoBERTa (Pre-trained)": [pre_trained_drb['f1'], pre_trained_drb['accuracy'], pre_trained_drb['weighted f1']],
    "RoBERTa (Pre-trained)": [pre_trained_rb['f1'], pre_trained_rb['accuracy'], pre_trained_rb['weighted f1']],
    "DistilBERT (Fine-tuned)": [ft_db_results['eval_f1'], ft_db_results['eval_accuracy'], ft_db_results['eval_weighted f1']],
    "RoBERTa (Fine-tuned)": [ft_rb_results['eval_f1'], ft_rb_results['eval_accuracy'], ft_rb_results['eval_weighted f1']]
}

comparison_df = pd.DataFrame(data, index=['Macro F1', 'Accuracy', 'Weighted F1'])

display(comparison_df)

# Plotting
comparison_df.T.plot(kind='bar', figsize=(10, 6))
plt.title("Model Performance Comparison")
plt.ylabel("Score")
plt.xticks(rotation=45)
plt.legend(loc='lower right')
plt.show()
```

[9/9 00:01]

[9/9 00:02]

	DistilRoBERTa (Pre-trained)	RoBERTa (Pre-trained)	DistilBERT (Fine-tuned)	RoBERTa (Fine-tuned)
Macro F1	0.890747	0.106576	0.183058	0.356798
Accuracy	0.911111	0.126667	0.441176	0.514706
Weighted F1	0.911611	0.090294	0.344530	0.478981

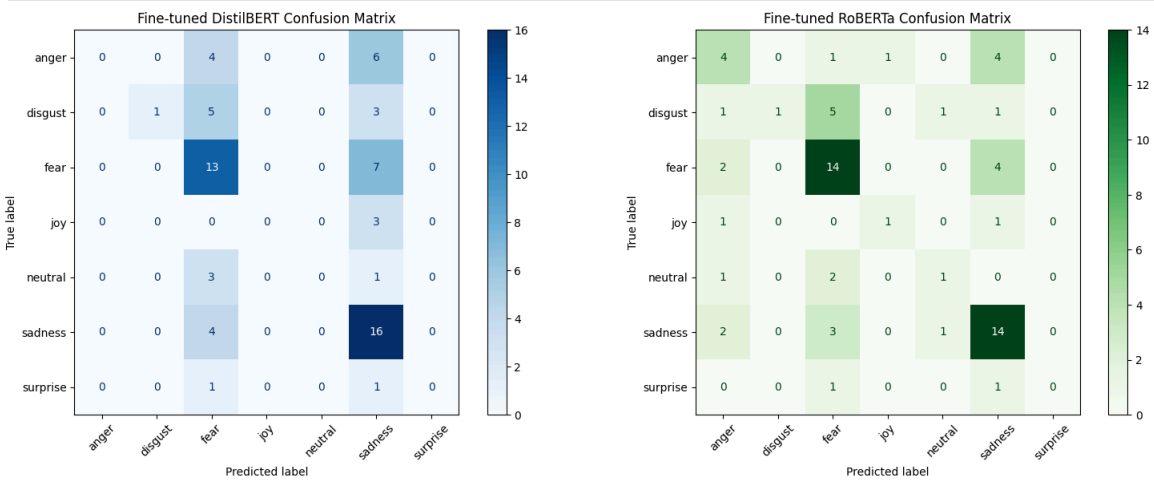


Confusion Matrices for Fine-Tuned Models

We'll now visualize where the fine-tuned models are succeeding or confusing specific poetic emotions.

In [31]:

```
import matplotlib.pyplot as plt
from sklearn.metrics import confusion_matrix, ConfusionMatrixDisplay

def get_predictions(trainer, dataset):
    predictions = trainer.predict(dataset)
    return np.argmax(predictions.predictions, axis=-1)

# Get predictions
ft_db_preds = get_predictions(trainer_db, val_db)
ft_rb_preds = get_predictions(trainer_rb, val_rb)

# Get true labels from validation set
y_true = val_df['label_num'].values

# Plotting
fig, (ax1, ax2) = plt.subplots(1, 2, figsize=(16, 6))

# DistilBERT CM
cm_db = confusion_matrix(y_true, ft_db_preds)
disp_db = ConfusionMatrixDisplay(confusion_matrix=cm_db, display_labels=label_list)
disp_db.plot(ax=ax1, cmap='Blues', xticks_rotation=45)
ax1.set_title("Fine-tuned DistilBERT Confusion Matrix")

# RoBERTa CM
cm_rb = confusion_matrix(y_true, ft_rb_preds)
disp_rb = ConfusionMatrixDisplay(confusion_matrix=cm_rb, display_labels=label_list)
disp_rb.plot(ax=ax2, cmap='Greens', xticks_rotation=45)
ax2.set_title("Fine-tuned RoBERTa Confusion Matrix")

plt.tight_layout()
plt.show()
```



Part 3: Augmenting Data

Because our dataset only contains 450 poems, we will augment the dataset, comparing how the models perform on Synonym augmentation versus Back Translation augmentation.

1. Wordnet Synonynm Augmentation

In [55]:
```
!pip install nlpaug nltk
import nltk
nltk.download('wordnet')
nltk.download('averaged_perceptron_tagger')
nltk.download('averaged_perceptron_tagger_eng')
nltk.download('omw-1.4')

import nlpaug.augmenter.word as naw

augmenter = naw.SynonymAug(aug_src='wordnet')
```

```
Collecting nlpaug
  Downloading nlpaug-1.1.11-py3-none-any.whl.metadata (14 kB)
Requirement already satisfied: nltk in /usr/local/lib/python3.12/dist-packages (3.9.1)
Requirement already satisfied: numpy>=1.16.2 in /usr/local/lib/python3.12/dist-packages (from nlpaug) (2.0.2)
Requirement already satisfied: pandas>=1.2.0 in /usr/local/lib/python3.12/dist-packages (from nlpaug) (2.2.2)
Requirement already satisfied: requests>=2.22.0 in /usr/local/lib/python3.12/dist-packages (from nlpaug) (2.32.4)
Requirement already satisfied: gdown>=4.0.0 in /usr/local/lib/python3.12/dist-packages (from nlpaug) (5.2.2)
Requirement already satisfied: click in /usr/local/lib/python3.12/dist-packages (from nltk) (8.3.3)
Requirement already satisfied: joblib in /usr/local/lib/python3.12/dist-packages (from nltk) (1.5.3)
Requirement already satisfied: regex>=2021.8.3 in /usr/local/lib/python3.12/dist-packages (from nltk) (2025.11.3)
Requirement already satisfied: tqdm in /usr/local/lib/python3.12/dist-packages (from nltk) (4.67.3)
Requirement already satisfied: beautifulsoup4 in /usr/local/lib/python3.12/dist-packages (from gdown>=4.0.0->nlpaug) (4.13.5)
Requirement already satisfied: filelock in /usr/local/lib/python3.12/dist-packages (from gdown>=4.0.0->nlpaug) (3.29.0)
Requirement already satisfied: python-dateutil>=2.8.2 in /usr/local/lib/python3.12/dist-packages (from pandas>=1.2.0->nlpaug) (2.9.0.post0)
Requirement already satisfied: pytz>=2020.1 in /usr/local/lib/python3.12/dist-packages (from pandas>=1.2.0->nlpaug) (2025.2)
Requirement already satisfied: tzdata>=2022.7 in /usr/local/lib/python3.12/dist-packages (from pandas>=1.2.0->nlpaug) (2026.1)
Requirement already satisfied: charset_normalizer<4,>=2 in /usr/local/lib/python3.12/dist-packages (from requests>=2.22.0->nlpaug) (3.4.7)
Requirement already satisfied: idna<4,>=2.5 in /usr/local/lib/python3.12/dist-packages (from requests>=2.22.0->nlpaug) (3.13)
Requirement already satisfied: urllib3<3,>=1.21.1 in /usr/local/lib/python3.12/dist-packages (from requests>=2.22.0->nlpaug) (2.5.0)
Requirement already satisfied: certifi>=2017.4.17 in /usr/local/lib/python3.12/dist-packages (from requests>=2.22.0->nlpaug) (2026.4.22)
Requirement already satisfied: six>=1.5 in /usr/local/lib/python3.12/dist-packages (from python-dateutil>=2.8.2->pandas>=1.2.0->nlpaug) (1.17.0)
Requirement already satisfied: soupsieve>1.2 in /usr/local/lib/python3.12/dist-packages (from beautifulsoup4->gdown>=4.0.0->nlpaug) (2.8.3)
Requirement already satisfied: typing-extensions>=4.0.0 in /usr/local/lib/python3.12/dist-packages (from beautifulsoup4->gdown>=4.0.0->nlpaug) (4.15.0)
Requirement already satisfied: PySocks!=1.5.7,>=1.5.6 in /usr/local/lib/python3.12/dist-packages (from requests[socks]->gdown>=4.0.0->nlpaug) (1.7.1)
Downloading nlpaug-1.1.11-py3-none-any.whl (410 kB)
   ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 410.5/410.5 kB 29.4 MB/s eta 0:00:00
Installing collected packages: nlpaug
Successfully installed nlpaug-1.1.11
```

```
[nltk_data] Downloading package wordnet to /root/nltk_data...
[nltk_data] Downloading package averaged_perceptron_tagger to
[nltk_data]     /root/nltk_data...
[nltk_data]   Unzipping taggers/averaged_perceptron_tagger.zip.
[nltk_data] Downloading package averaged_perceptron_tagger_eng to
[nltk_data]     /root/nltk_data...
[nltk_data]   Unzipping taggers/averaged_perceptron_tagger_eng.zip.
[nltk_data] Downloading package omw-1.4 to /root/nltk_data...
```

In [56]:
```
from tqdm import tqdm

def augment_dataframe(df, augmenter, text_col='poem content'):
    augmented_rows = []
    for _, row in tqdm(df.iterrows(), total=len(df)):
        augmented_text = augmenter.augment(row[text_col])[0]
        augmented_rows.append({
            'poem content': augmented_text,
            'label_num': row['label_num']
        })
    augmented_df = pds.DataFrame(augmented_rows)
    return pds.concat([df, augmented_df], ignore_index=True)

train_df_augmented = augment_dataframe(train_df, augmenter)

train_dataset = Dataset.from_pandas(train_df_augmented[['poem content', 'label_num']])
val_dataset = Dataset.from_pandas(val_df[['poem content', 'label_num']])
```

```
100%|██████████| 382/382 [00:03<00:00, 120.16it/s]
```

In [57]:
```
from transformers import AutoTokenizer, AutoModelForSequenceClassification, TrainingArguments, Trainer

def get_tokenized_dataset(model_name, dataset):
    tokenizer = AutoTokenizer.from_pretrained(model_name)
    def tokenize_func(examples):
        result = tokenizer(examples['poem content'], padding='max_length', truncation=True)
        result['labels'] = examples['label_num']
        return result
    return dataset.map(tokenize_func, batched=True), tokenizer

# Setup for DistilBERT
db_model_name = "distilbert-base-uncased"
train_db, db_tokenizer = get_tokenized_dataset(db_model_name, train_dataset)
val_db, _ = get_tokenized_dataset(db_model_name, val_dataset)

# Setup for RoBERTa
rb_model_name = "roberta-base"
train_rb, rb_tokenizer = get_tokenized_dataset(rb_model_name, train_dataset)
val_rb, _ = get_tokenized_dataset(rb_model_name, val_dataset)
```

```
Map:   0%|          | 0/764 [00:00<?, ? examples/s]
Map:   0%|          | 0/68 [00:00<?, ? examples/s]
Map:   0%|          | 0/764 [00:00<?, ? examples/s]
Map:   0%|          | 0/68 [00:00<?, ? examples/s]
```

In [59]:
```
def compute_metrics(eval_pred):
    logits, labels = eval_pred
    predictions = np.argmax(logits, axis=-1)
    return pred_metrics(le.inverse_transform(labels), le.inverse_transform(predictions))

training_args = TrainingArguments(
    output_dir="./results",
    eval_strategy="epoch",
    learning_rate=2e-5,
    per_device_train_batch_size=16,
    num_train_epochs=10,
    weight_decay=0.05,
)

# Initialize DistilBERT for Fine-Tuning
wordnet_ft_distilbert = AutoModelForSequenceClassification.from_pretrained(db_model_name, num_labels=len(label_list))

wordnet_trainer_db = Trainer(
    model=wordnet_ft_distilbert,
    args=training_args,
    train_dataset=train_db,
    eval_dataset=val_db,
```

```
    compute_metrics=compute_metrics,
)

print("Starting Fine-tuning for DistilBERT...")
wordnet_trainer_db.train()
```

```
Loading weights:   0%|          | 0/100 [00:00<?, ?it/s]
DistilBertForSequenceClassification LOAD REPORT from: distilbert-base-uncased
Key                     | Status     |
------------------------+------------+-
vocab_layer_norm.bias   | UNEXPECTED |
vocab_projector.bias    | UNEXPECTED |
vocab_layer_norm.weight | UNEXPECTED |
vocab_transform.bias    | UNEXPECTED |
vocab_transform.weight  | UNEXPECTED |
classifier.weight       | MISSING    |
pre_classifier.weight   | MISSING    |
pre_classifier.bias     | MISSING    |
classifier.bias         | MISSING    |

Notes:
- UNEXPECTED    :can be ignored when loading from different task/architecture; not ok if you expect identical arch.
- MISSING       :those params were newly initialized because missing from the checkpoint. Consider training on your downstream task.
Starting Fine-tuning for DistilBERT...
```

[480/480 02:39, Epoch 10/10]

Epoch	Training Loss	Validation Loss	F1	Accuracy	Precision	Recall	Weighted f1
1	No log	1.594079	0.153571	0.426471	0.123469	0.207143	0.316176
2	No log	1.556409	0.157483	0.441176	0.127159	0.214286	0.324231
3	No log	1.506322	0.177482	0.426471	0.262690	0.215873	0.333051
4	No log	1.510956	0.250943	0.500000	0.358142	0.273016	0.429146
5	No log	1.676264	0.188906	0.411765	0.315476	0.215873	0.335225
6	No log	1.534797	0.313577	0.500000	0.565637	0.306349	0.439228
7	No log	1.581208	0.382440	0.529412	0.575893	0.363492	0.486232
8	No log	1.671353	0.412860	0.529412	0.661565	0.384921	0.490325
9	No log	1.699850	0.420238	0.544118	0.610712	0.399206	0.504185
10	No log	1.725551	0.374137	0.514706	0.602041	0.349206	0.471472

```
Writing model shards:   0%|          | 0/1 [00:00<?, ?it/s]
```

Out[59]:
```
TrainOutput(global_step=480, training_loss=0.7334388097127279, metrics={'train_runtime': 160.0286, 'train_samples_per_second': 47.741, 'train_steps_per_second': 2.999, 'total_flos': 1012141168189440.0, 'train_loss': 0.7334388097127279, 'epoch': 10.0})
```

In [60]:
```
wordnet_ft_roberta = AutoModelForSequenceClassification.from_pretrained(rb_model_name, num_labels=len(label_list))

wordnet_trainer_rb = Trainer(
    model=wordnet_ft_roberta,
    args=training_args,
    train_dataset=train_rb,
    eval_dataset=val_rb,
    compute_metrics=compute_metrics,
)

print("Starting Fine-tuning for RoBERTa...")
wordnet_trainer_rb.train()
```

```
Loading weights:   0%|          | 0/197 [00:00<?, ?it/s]
RobertaForSequenceClassification LOAD REPORT from: roberta-base
Key                             | Status     |
--------------------------------+------------+-
lm_head.layer_norm.weight       | UNEXPECTED |
roberta.embeddings.position_ids | UNEXPECTED |
lm_head.bias                    | UNEXPECTED |
lm_head.layer_norm.bias         | UNEXPECTED |
lm_head.dense.weight            | UNEXPECTED |
lm_head.dense.bias              | UNEXPECTED |
classifier.dense.bias           | MISSING    |
classifier.out_proj.bias        | MISSING    |
classifier.dense.weight         | MISSING    |
classifier.out_proj.weight      | MISSING    |

Notes:
- UNEXPECTED    :can be ignored when loading from different task/architecture; not ok if you expect identical arch.
- MISSING       :those params were newly initialized because missing from the checkpoint. Consider training on your downstream task.
Starting Fine-tuning for RoBERTa...
```

[480/480 05:25, Epoch 10/10]

Epoch	Training Loss	Validation Loss	F1	Accuracy	Precision	Recall	Weighted f1
1	No log	1.633111	0.157823	0.441176	0.128372	0.214286	0.324930
2	No log	1.455457	0.151566	0.411765	0.122186	0.200000	0.312048
3	No log	1.339605	0.339438	0.455882	0.433571	0.342063	0.450918
4	No log	1.332639	0.347533	0.544118	0.447527	0.389683	0.499332
5	No log	1.371886	0.397561	0.544118	0.555736	0.379365	0.530108
6	No log	1.318851	0.412920	0.588235	0.500411	0.407937	0.558953
7	No log	1.432822	0.470938	0.573529	0.572234	0.436508	0.569348
8	No log	1.376943	0.569952	0.632353	0.710031	0.519841	0.650739
9	No log	1.507017	0.517529	0.632353	0.733242	0.484921	0.597691
10	No log	1.456141	0.572568	0.647059	0.741771	0.518254	0.644235

```
Writing model shards:   0%|          | 0/1 [00:00<?, ?it/s]
```

Out[60]:
```
TrainOutput(global_step=480, training_loss=0.6362691243489583, metrics={'train_runtime': 326.2934, 'train_samples_per_second': 23.415, 'train_steps_per_second': 1.471, 'total_flos': 2010258705408000.0, 'train_loss': 0.6362691243489583, 'epoch': 10.0})
```

2. Back Translantion

In [61]:
```
augmenter = naw.BackTranslationAug(
    from_model_name='Helsinki-NLP/opus-mt-en-fr',
    to_model_name='Helsinki-NLP/opus-mt-fr-en',
    batch_size=8
)

train_df_augmented = augment_dataframe(train_df, augmenter)

train_dataset = Dataset.from_pandas(train_df_augmented[['poem content', 'label_num']])
```

```
val_dataset = Dataset.from_pandas(val_df[['poem content', 'label_num']])

train_df_augmented.to_csv('augmented_train.csv', index=False)
```

```
config.json: 0.00B [00:00, ?B/s]
pytorch_model.bin:   0%|          | 0.00/301M [00:00<?, ?B/s]
Loading weights:   0%|          | 0/258 [00:00<?, ?it/s]
The tied weights mapping and config for this model specifies to tie model.shared.weight to model.decoder.embed_tokens.weight, but both are present in the checkpoints, so we will NOT tie them. You should update the config with `tie_word_embeddings=False` to silence this warning
The tied weights mapping and config for this model specifies to tie model.shared.weight to model.encoder.embed_tokens.weight, but both are present in the checkpoints, so we will NOT tie them. You should update the config with `tie_word_embeddings=False` to silence this warning
model.safetensors:   0%|          | 0.00/301M [00:00<?, ?B/s]
generation_config.json:   0%|          | 0.00/293 [00:00<?, ?B/s]
config.json: 0.00B [00:00, ?B/s]
model.safetensors:   0%|          | 0.00/301M [00:00<?, ?B/s]
Loading weights:   0%|          | 0/256 [00:00<?, ?it/s]
generation_config.json:   0%|          | 0.00/293 [00:00<?, ?B/s]
tokenizer_config.json:   0%|          | 0.00/42.0 [00:00<?, ?B/s]
source.spm:   0%|          | 0.00/778k [00:00<?, ?B/s]
target.spm:   0%|          | 0.00/802k [00:00<?, ?B/s]
vocab.json: 0.00B [00:00, ?B/s]
/usr/local/lib/python3.12/dist-packages/transformers/models/marian/tokenization_marian.py:176: UserWarning: Recommended: pip install sacremoses.
  warnings.warn("Recommended: pip install sacremoses.")
tokenizer_config.json:   0%|          | 0.00/42.0 [00:00<?, ?B/s]
source.spm:   0%|          | 0.00/802k [00:00<?, ?B/s]
target.spm:   0%|          | 0.00/778k [00:00<?, ?B/s]
vocab.json: 0.00B [00:00, ?B/s]
100%|██████████| 382/382 [47:29<00:00,  7.46s/it]
```

In [62]:

```
def compute_metrics(eval_pred):
    logits, labels = eval_pred
    predictions = np.argmax(logits, axis=-1)
    return pred_metrics(le.inverse_transform(labels), le.inverse_transform(predictions))

training_args = TrainingArguments(
    output_dir="./results",
    eval_strategy="epoch",
    learning_rate=2e-5,
    per_device_train_batch_size=16,
    num_train_epochs=10,
    weight_decay=0.05,
)

# Initialize DistilBERT for Fine-Tuning
bt_ft_distilbert = AutoModelForSequenceClassification.from_pretrained(db_model_name, num_labels=len(label_list))

bt_trainer_db = Trainer(
    model=bt_ft_distilbert,
    args=training_args,
    train_dataset=train_db,
    eval_dataset=val_db,
    compute_metrics=compute_metrics,
)

print("Starting Fine-tuning for DistilBERT...")
bt_trainer_db.train()
```

```
Loading weights:   0%|          | 0/100 [00:00<?, ?it/s]
DistilBertForSequenceClassification LOAD REPORT from: distilbert-base-uncased
Key                     | Status     |
------------------------+------------+-
vocab_layer_norm.bias   | UNEXPECTED |
vocab_projector.bias    | UNEXPECTED |
vocab_layer_norm.weight | UNEXPECTED |
vocab_transform.bias    | UNEXPECTED |
vocab_transform.weight  | UNEXPECTED |
classifier.weight       | MISSING    |
pre_classifier.weight   | MISSING    |
pre_classifier.bias     | MISSING    |
classifier.bias         | MISSING    |

Notes:
- UNEXPECTED    :can be ignored when loading from different task/architecture; not ok if you expect identical arch.
- MISSING       :those params were newly initialized because missing from the checkpoint. Consider training on your downstream task.
Starting Fine-tuning for DistilBERT...
```

[480/480 02:38, Epoch 10/10]

Epoch	Training Loss	Validation Loss	F1	Accuracy	Precision	Recall	Weighted f1
1	No log	1.578109	0.162092	0.455882	0.130560	0.221429	0.333718
2	No log	1.501644	0.158556	0.441176	0.125992	0.214286	0.326438
3	No log	1.521881	0.171694	0.426471	0.192177	0.214286	0.328978
4	No log	1.502210	0.163384	0.397059	0.161797	0.200000	0.313754
5	No log	1.778940	0.175159	0.382353	0.301190	0.201587	0.305645
6	No log	1.534716	0.271005	0.441176	0.283956	0.277778	0.414363
7	No log	1.572293	0.312884	0.441176	0.355714	0.299206	0.433198
8	No log	1.581108	0.325122	0.470588	0.351648	0.320635	0.451186
9	No log	1.657076	0.326478	0.455882	0.434524	0.311905	0.444559
10	No log	1.654623	0.325754	0.455882	0.432889	0.311905	0.440616

```
Writing model shards:   0%|          | 0/1 [00:00<?, ?it/s]
```

Out[62]:

```
TrainOutput(global_step=480, training_loss=0.8006631215413411, metrics={'train_runtime': 158.9853, 'train_samples_per_second': 48.055, 'train_steps_per_second': 3.019, 'total_flos': 1012141168189440.0, 'train_loss': 0.8006631215413411, 'epoch': 10.0})
```

In [63]:

```
bt_ft_roberta = AutoModelForSequenceClassification.from_pretrained(rb_model_name, num_labels=len(label_list))

bt_trainer_rb = Trainer(
    model=bt_ft_roberta,
    args=training_args,
    train_dataset=train_rb,
    eval_dataset=val_rb,
    compute_metrics=compute_metrics,
)

print("Starting Fine-tuning for RoBERTa...")
bt_trainer_rb.train()
```

```
Loading weights:   0%|          | 0/197 [00:00<?, ?it/s]
```

```
RobertaForSequenceClassification LOAD REPORT from: roberta-base
Key                             | Status     |
--------------------------------+------------+-
lm_head.layer_norm.weight       | UNEXPECTED |
roberta.embeddings.position_ids | UNEXPECTED |
lm_head.bias                    | UNEXPECTED |
lm_head.layer_norm.bias         | UNEXPECTED |
lm_head.dense.weight            | UNEXPECTED |
lm_head.dense.bias              | UNEXPECTED |
classifier.dense.bias           | MISSING    |
classifier.out_proj.bias        | MISSING    |
classifier.dense.weight         | MISSING    |
classifier.out_proj.weight      | MISSING    |

Notes:
- UNEXPECTED    :can be ignored when loading from different task/architecture; not ok if you expect identical arch.
- MISSING       :those params were newly initialized because missing from the checkpoint. Consider training on your downstream task.
Starting Fine-tuning for RoBERTa...
```

[480/480 05:26, Epoch 10/10]

Epoch	Training Loss	Validation Loss	F1	Accuracy	Precision	Recall	Weighted f1
1	No log	1.633814	0.152574	0.426471	0.122726	0.207143	0.314122
2	No log	1.452571	0.151566	0.411765	0.122186	0.200000	0.312048
3	No log	1.275057	0.369361	0.500000	0.447661	0.377778	0.492241
4	No log	1.276486	0.346804	0.544118	0.444112	0.389683	0.491250
5	No log	1.350168	0.377368	0.544118	0.554022	0.370635	0.511442
6	No log	1.309211	0.541052	0.617647	0.734921	0.507937	0.608554
7	No log	1.464477	0.486779	0.588235	0.656494	0.438095	0.567237
8	No log	1.362663	0.563664	0.632353	0.712654	0.512698	0.641443
9	No log	1.562014	0.497232	0.602941	0.717757	0.463492	0.568399
10	No log	1.505902	0.553487	0.632353	0.738370	0.502381	0.627894

```
Writing model shards:   0%|          | 0/1 [00:00<?, ?it/s]
```

Out[63]:
```
TrainOutput(global_step=480, training_loss=0.6389704386393229, metrics={'train_runtime': 327.1606, 'train_samples_per_second': 23.352, 'train_steps_per_second': 1.467, 'total_flos': 2010258705408000.0, 'train_loss': 0.6389704386393229, 'epoch': 10.0})
```

3. Synthetic Dataset

In [32]:
```python
from datasets import load_dataset
import pandas as pd
from tqdm import tqdm

ds = load_dataset("biglam/gutenberg-poetry-corpus")

poems_df = ds['train'].to_pandas()

print(poems_df.columns.tolist())
print(poems_df.head(2))
```

```
README.md: 0.00B [00:00, ?B/s]
dataset_infos.json:   0%|          | 0.00/903 [00:00<?, ?B/s]
data/train-00000-of-00001-fa9fb9e1f16eed(…):   0%|          | 0.00/91.1M [00:00<?, ?B/s]
Generating train split:   0%|          | 0/3085117 [00:00<?, ? examples/s]
['line', 'gutenberg_id']
                                                line  gutenberg_id
0  The Song of Hiawatha is based on the legends a...            19
1  many North American Indian tribes, but especia...            19
```

In [33]:
```python
poems_df = poems_df.sample(n=100000, random_state=42).reset_index(drop=True)

poems_df = poems_df.rename(columns={'line': 'poem content'})

poems_df = poems_df.dropna(subset=['poem content'])

def label_poems(df, classifier, text_col='poem content', batch_size=16):
    results = []
    poems = df[text_col].tolist()

    for i in tqdm(range(0, len(poems), batch_size)):
        batch = poems[i:i+batch_size]
        predictions = classifier(batch, truncation=True, max_length=512)

        for poem, pred in zip(batch, predictions):
            results.append({
                'poem content': poem,
                'label': pred['label'],
                'confidence': round(pred['score'], 4)
            })

    return pd.DataFrame(results)

labelled_df = label_poems(poems_df, DistilRoBERTa_classifier)

high_conf_df = labelled_df[labelled_df['confidence'] >= 0.85]

labelled_df.to_csv('synthetic_labelled_gutenberg.csv', index=False)
high_conf_df.to_csv('synthetic_labelled_gutenberg_highconf.csv', index=False)

print(f"Total labelled: {len(labelled_df)}")
print(f"High confidence: {len(high_conf_df)}")
print(high_conf_df['label'].value_counts())
```

```
100%|██████████| 6250/6250 [25:39<00:00,  4.06it/s]
Total labelled: 100000
High confidence: 23192
label
neutral     9457
joy         3456
sadness     3224
fear        2470
disgust     2045
anger       1924
surprise     616
Name: count, dtype: int64
```

In [34]:
```python
from datasets import Dataset
import numpy as np


all_labels = np.union1d(high_conf_df['label'].unique(), newData['label'].unique())
le.fit(all_labels)
```

```
high_conf_df['label_num'] = le.transform(high_conf_df['label'])
newData['label_num'] = le.transform(newData['label'])

synthetic_dataset = Dataset.from_pandas(high_conf_df[['poem content', 'label_num']])
poem_dataset = Dataset.from_pandas(newData[['poem content', 'label_num']])

train_dataset = synthetic_dataset
val_dataset = poem_dataset
```

```
/tmp/ipykernel_18276/518495558.py:8: SettingWithCopyWarning:
A value is trying to be set on a copy of a slice from a DataFrame.
Try using .loc[row_indexer,col_indexer] = value instead

See the caveats in the documentation: https://pandas.pydata.org/pandas-docs/stable/user_guide/indexing.html#returning-a-view-versus-a-copy
  high_conf_df['label_num'] = le.transform(high_conf_df['label'])
```

In [35]:
```
from transformers import AutoTokenizer, AutoModelForSequenceClassification, TrainingArguments, Trainer

def get_tokenized_dataset(model_name, dataset):
    tokenizer = AutoTokenizer.from_pretrained(model_name)
    def tokenize_func(examples):
        result = tokenizer(examples['poem content'], padding='max_length', truncation=True)
        result['labels'] = examples['label_num']
        return result
    return dataset.map(tokenize_func, batched=True), tokenizer

# Setup for DistilBERT
db_model_name = "distilbert-base-uncased"
train_db, db_tokenizer = get_tokenized_dataset(db_model_name, train_dataset)
val_db, _ = get_tokenized_dataset(db_model_name, val_dataset)

# Setup for RoBERTa
rb_model_name = "roberta-base"
train_rb, rb_tokenizer = get_tokenized_dataset(rb_model_name, train_dataset)
val_rb, _ = get_tokenized_dataset(rb_model_name, val_dataset)
```

```
Map:   0%|          | 0/23192 [00:00<?, ? examples/s]
Map:   0%|          | 0/450 [00:00<?, ? examples/s]
Map:   0%|          | 0/23192 [00:00<?, ? examples/s]
Map:   0%|          | 0/450 [00:00<?, ? examples/s]
```

In [37]:
```
def compute_metrics(eval_pred):
    logits, labels = eval_pred
    predictions = np.argmax(logits, axis=-1)
    return pred_metrics(le.inverse_transform(labels), le.inverse_transform(predictions))

training_args = TrainingArguments(
    output_dir="./results",
    eval_strategy="epoch",
    save_strategy="epoch",
    learning_rate=2e-5,
    per_device_train_batch_size=16,
    num_train_epochs=3,
    weight_decay=0.05,
    load_best_model_at_end=True,
    metric_for_best_model="f1",
    greater_is_better=True,
)

# Initialize DistilBERT for Fine-Tuning
ft_distilbert = AutoModelForSequenceClassification.from_pretrained(db_model_name, num_labels=len(label_list))

trainer_db = Trainer(
    model=ft_distilbert,
    args=training_args,
    train_dataset=train_db,
    eval_dataset=val_db,
    compute_metrics=compute_metrics,
)

print("Starting Fine-tuning for DistilBERT...")
trainer_db.train()
```

```
Loading weights:   0%|          | 0/100 [00:00<?, ?it/s]
DistilBertForSequenceClassification LOAD REPORT from: distilbert-base-uncased
Key                     | Status     |
------------------------+------------+-
vocab_layer_norm.bias   | UNEXPECTED |
vocab_projector.bias    | UNEXPECTED |
vocab_layer_norm.weight | UNEXPECTED |
vocab_transform.bias    | UNEXPECTED |
vocab_transform.weight  | UNEXPECTED |
classifier.weight       | MISSING    |
pre_classifier.weight   | MISSING    |
pre_classifier.bias     | MISSING    |
classifier.bias         | MISSING    |

Notes:
- UNEXPECTED    :can be ignored when loading from different task/architecture; not ok if you expect identical arch.
- MISSING       :those params were newly initialized because missing from the checkpoint. Consider training on your downstream task.
Starting Fine-tuning for DistilBERT...
```

[4350/4350 23:21, Epoch 3/3]

Epoch	Training Loss	Validation Loss	F1	Accuracy	Precision	Recall	Weighted f1
1	0.437960	1.174370	0.514168	0.577778	0.524338	0.517993	0.577885
2	0.205019	1.598133	0.532089	0.566667	0.528905	0.568116	0.570625
3	0.103063	1.710956	0.542046	0.586667	0.540346	0.561469	0.590092

```
Writing model shards:   0%|          | 0/1 [00:00<?, ?it/s]
Writing model shards:   0%|          | 0/1 [00:00<?, ?it/s]
Writing model shards:   0%|          | 0/1 [00:00<?, ?it/s]
```

```
There were missing keys in the checkpoint model loaded: ['distilbert.embeddings.LayerNorm.weight', 'distilbert.embeddings.LayerNorm.bias'].
There were unexpected keys in the checkpoint model loaded: ['distilbert.embeddings.LayerNorm.beta', 'distilbert.embeddings.LayerNorm.gamma'].
```

Out[37]:
```
TrainOutput(global_step=4350, training_loss=0.29677229958019036, metrics={'train_runtime': 1401.4481, 'train_samples_per_second': 49.646, 'train_steps_per_second': 3.104, 'total_flos': 9217373549469696.0, 'train_loss': 0.29677229958019036, 'epoch': 3.0})
```

In [40]:
```
ft_roberta = AutoModelForSequenceClassification.from_pretrained(rb_model_name, num_labels=len(label_list))

trainer_rb = Trainer(
    model=ft_roberta,
    args=training_args,
    train_dataset=train_rb,
    eval_dataset=val_rb,
    compute_metrics=compute_metrics,
)
```

```
print("Starting Fine-tuning for RoBERTa...")
trainer_rb.train()
```

```
Loading weights:   0%|          | 0/197 [00:00<?, ?it/s]
RobertaForSequenceClassification LOAD REPORT from: roberta-base
Key                             | Status     |
--------------------------------+------------+-
lm_head.layer_norm.weight       | UNEXPECTED |
roberta.embeddings.position_ids | UNEXPECTED |
lm_head.bias                    | UNEXPECTED |
lm_head.layer_norm.bias         | UNEXPECTED |
lm_head.dense.weight            | UNEXPECTED |
lm_head.dense.bias              | UNEXPECTED |
classifier.dense.bias           | MISSING    |
classifier.out_proj.bias        | MISSING    |
classifier.dense.weight         | MISSING    |
classifier.out_proj.weight      | MISSING    |

Notes:
- UNEXPECTED    :can be ignored when loading from different task/architecture; not ok if you expect identical arch.
- MISSING       :those params were newly initialized because missing from the checkpoint. Consider training on your downstream task.
Starting Fine-tuning for RoBERTa...
```

[4350/4350 46:50, Epoch 3/3]

Epoch	Training Loss	Validation Loss	F1	Accuracy	Precision	Recall	Weighted f1
1	0.334611	1.541601	0.511178	0.575556	0.583578	0.531888	0.560448
2	0.163618	1.724594	0.546468	0.615556	0.562202	0.591288	0.622622
3	0.084640	1.998684	0.529166	0.611111	0.568270	0.552327	0.604410

```
Writing model shards:   0%|          | 0/1 [00:00<?, ?it/s]
Writing model shards:   0%|          | 0/1 [00:00<?, ?it/s]
Writing model shards:   0%|          | 0/1 [00:00<?, ?it/s]
There were missing keys in the checkpoint model loaded: ['roberta.embeddings.LayerNorm.weight', 'roberta.embeddings.LayerNorm.bias', 'roberta.encoder.layer.0.attention.output.LayerNorm.weight', 'roberta.encoder.layer.0.attention.output.LayerNorm.bias', 'roberta.encoder.layer.0.output.LayerNorm.weight', 'roberta.encoder.layer.0.output.LayerNorm.bias',
'roberta.encoder.layer.1.attention.output.LayerNorm.weight', 'roberta.encoder.layer.1.attention.output.LayerNorm.bias', 'roberta.encoder.layer.1.output.LayerNorm.weight', 'roberta.encoder.layer.1.output.LayerNorm.bias', 'roberta.encoder.layer.2.attention.output.LayerNorm.weight', 'roberta.encoder.layer.2.attention.output.LayerNorm.bias', 'roberta.enc
oder.layer.2.output.LayerNorm.weight', 'roberta.encoder.layer.2.output.LayerNorm.bias', 'roberta.encoder.layer.3.attention.output.LayerNorm.weight', 'roberta.encoder.layer.3.attention.output.LayerNorm.bias', 'roberta.encoder.layer.3.output.LayerNorm.weight', 'roberta.encoder.layer.3.output.LayerNorm.bias', 'roberta.encoder.layer.4.attention.output.La
yerNorm.weight', 'roberta.encoder.layer.4.attention.output.LayerNorm.bias', 'roberta.encoder.layer.4.output.LayerNorm.weight', 'roberta.encoder.layer.4.output.LayerNorm.bias', 'roberta.encoder.layer.5.attention.output.LayerNorm.weight', 'roberta.encoder.layer.5.attention.output.LayerNorm.bias', 'roberta.encoder.layer.5.output.LayerNorm.weight', 'robe
rta.encoder.layer.5.output.LayerNorm.bias', 'roberta.encoder.layer.6.attention.output.LayerNorm.weight', 'roberta.encoder.layer.6.attention.output.LayerNorm.bias', 'roberta.encoder.layer.6.output.LayerNorm.weight', 'roberta.encoder.layer.6.output.LayerNorm.bias', 'roberta.encoder.layer.7.attention.output.LayerNorm.weight', 'roberta.encoder.layer.7.at
tention.output.LayerNorm.bias', 'roberta.encoder.layer.7.output.LayerNorm.weight', 'roberta.encoder.layer.7.output.LayerNorm.bias', 'roberta.encoder.layer.8.attention.output.LayerNorm.weight', 'roberta.encoder.layer.8.attention.output.LayerNorm.bias', 'roberta.encoder.layer.8.output.LayerNorm.weight', 'roberta.encoder.layer.8.output.LayerNorm.bias',
'roberta.encoder.layer.9.attention.output.LayerNorm.weight', 'roberta.encoder.layer.9.attention.output.LayerNorm.bias', 'roberta.encoder.layer.9.output.LayerNorm.weight', 'roberta.encoder.layer.9.output.LayerNorm.bias', 'roberta.encoder.layer.10.attention.output.LayerNorm.weight', 'roberta.encoder.layer.10.attention.output.LayerNorm.bias', 'roberta.e
ncoder.layer.10.output.LayerNorm.weight', 'roberta.encoder.layer.10.output.LayerNorm.bias', 'roberta.encoder.layer.11.attention.output.LayerNorm.weight', 'roberta.encoder.layer.11.attention.output.LayerNorm.bias', 'roberta.encoder.layer.11.output.LayerNorm.weight', 'roberta.encoder.layer.11.output.LayerNorm.bias'].
There were unexpected keys in the checkpoint model loaded: ['roberta.embeddings.LayerNorm.beta', 'roberta.embeddings.LayerNorm.gamma', 'roberta.encoder.layer.0.attention.output.LayerNorm.beta', 'roberta.encoder.layer.0.attention.output.LayerNorm.gamma', 'roberta.encoder.layer.0.output.LayerNorm.beta', 'roberta.encoder.layer.0.output.LayerNorm.gamma',
'roberta.encoder.layer.1.attention.output.LayerNorm.beta', 'roberta.encoder.layer.1.attention.output.LayerNorm.gamma', 'roberta.encoder.layer.1.output.LayerNorm.beta', 'roberta.encoder.layer.1.output.LayerNorm.gamma', 'roberta.encoder.layer.2.attention.output.LayerNorm.beta', 'roberta.encoder.layer.2.attention.output.LayerNorm.gamma', 'roberta.encode
r.layer.2.output.LayerNorm.beta', 'roberta.encoder.layer.2.output.LayerNorm.gamma', 'roberta.encoder.layer.3.attention.output.LayerNorm.beta', 'roberta.encoder.layer.3.attention.output.LayerNorm.gamma', 'roberta.encoder.layer.3.output.LayerNorm.beta', 'roberta.encoder.layer.3.output.LayerNorm.gamma', 'roberta.encoder.layer.4.attention.output.LayerNor
m.beta', 'roberta.encoder.layer.4.attention.output.LayerNorm.gamma', 'roberta.encoder.layer.4.output.LayerNorm.beta', 'roberta.encoder.layer.4.output.LayerNorm.gamma', 'roberta.encoder.layer.5.attention.output.LayerNorm.beta', 'roberta.encoder.layer.5.attention.output.LayerNorm.gamma', 'roberta.encoder.layer.5.output.LayerNorm.beta', 'roberta.encode
r.layer.5.output.LayerNorm.gamma', 'roberta.encoder.layer.6.attention.output.LayerNorm.beta', 'roberta.encoder.layer.6.attention.output.LayerNorm.gamma', 'roberta.encoder.layer.6.output.LayerNorm.beta', 'roberta.encoder.layer.6.output.LayerNorm.gamma', 'roberta.encoder.layer.7.attention.output.LayerNorm.beta', 'roberta.encoder.layer.7.attention.outpu
t.LayerNorm.gamma', 'roberta.encoder.layer.7.output.LayerNorm.beta', 'roberta.encoder.layer.7.output.LayerNorm.gamma', 'roberta.encoder.layer.8.attention.output.LayerNorm.beta', 'roberta.encoder.layer.8.attention.output.LayerNorm.gamma', 'roberta.encoder.layer.8.output.LayerNorm.beta', 'roberta.encoder.layer.8.output.LayerNorm.gamma', 'roberta.encode
r.layer.9.attention.output.LayerNorm.beta', 'roberta.encoder.layer.9.attention.output.LayerNorm.gamma', 'roberta.encoder.layer.9.output.LayerNorm.beta', 'roberta.encoder.layer.9.output.LayerNorm.gamma', 'roberta.encoder.layer.10.attention.output.LayerNorm.beta', 'roberta.encoder.layer.10.attention.output.LayerNorm.gamma', 'roberta.encoder.layer.10.ou
tput.LayerNorm.beta', 'roberta.encoder.layer.10.output.LayerNorm.gamma', 'roberta.encoder.layer.11.attention.output.LayerNorm.beta', 'roberta.encoder.layer.11.attention.output.LayerNorm.gamma', 'roberta.encoder.layer.11.output.LayerNorm.beta', 'roberta.encoder.layer.11.output.LayerNorm.gamma'].
```

Out[40]:
```
TrainOutput(global_step=4350, training_loss=0.23197233704314835, metrics={'train_runtime': 2811.62, 'train_samples_per_second': 24.746, 'train_steps_per_second': 1.547, 'total_flos': 1.83070366083072e+16, 'train_loss': 0.23197233704314835, 'epoch': 3.0})
```

In [42]:
```
# Save DistilBERT
trainer_db.save_model("./models/distilbert-finetuned")
db_tokenizer.save_pretrained("./models/distilbert-finetuned")

# Save RoBERTa
trainer_rb.save_model("./models/roberta-finetuned")
rb_tokenizer.save_pretrained("./models/roberta-finetuned")

# Save label encoder
import pickle
with open('./models/label_encoder.pkl', 'wb') as f:
    pickle.dump(le, f)

print("All models saved.")
```

```
Writing model shards:   0%|          | 0/1 [00:00<?, ?it/s]
All models saved.
```

In [45]:
```
from google.colab import files
import shutil

shutil.make_archive('distilbert-finetuned', 'zip', './models/distilbert-finetuned')
shutil.make_archive('roberta-finetuned', 'zip', './models/roberta-finetuned')
shutil.make_archive('label_encoder', 'zip', './models')

files.download('distilbert-finetuned.zip')
files.download('roberta-finetuned.zip')
```

4. Metrics and Comparisons

In [64]:
```
from transformers import AutoModelForSequenceClassification, TrainingArguments, Trainer

# Reload saved models
ft_distilbert = AutoModelForSequenceClassification.from_pretrained("./models/distilbert-finetuned")
ft_roberta = AutoModelForSequenceClassification.from_pretrained("./models/roberta-finetuned")

eval_args = TrainingArguments(
    output_dir="./eval_tmp",
    report_to="none",
)

trainer_db = Trainer(
    model=ft_distilbert,
    args=eval_args,
    eval_dataset=val_db,
    compute_metrics=compute_metrics,
)

trainer_rb = Trainer(
    model=ft_roberta,
    args=eval_args,
    eval_dataset=val_rb,
    compute_metrics=compute_metrics,
)


ft_db_results = trainer_db.evaluate()
ft_rb_results = trainer_rb.evaluate()
```

```
Loading weights:   0%|          | 0/104 [00:00<?, ?it/s]
Loading weights:   0%|          | 0/201 [00:00<?, ?it/s]
```

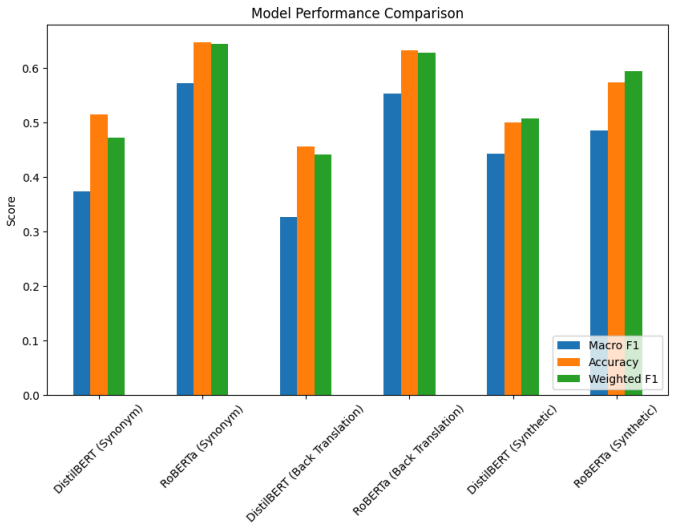
[9/9 00:00]

[9/9 00:00]

In [65]:
```python
import pandas as pd
import matplotlib.pyplot as plt

# 1. Synonym Augmentation
wordnet_db_results = wordnet_trainer_db.evaluate()
wordnet_rb_results = wordnet_trainer_rb.evaluate()

# 2. Backtranslation Augmentation
bt_db_results = bt_trainer_db.evaluate()
bt_rb_results = bt_trainer_rb.evaluate()

# 3. Synthetic Data
ft_db_results = trainer_db.evaluate()
ft_rb_results = trainer_rb.evaluate()

# Simplify keys for comparison
data = {
    "DistilBERT (Synonym)": [wordnet_db_results['eval_f1'], wordnet_db_results['eval_accuracy'], wordnet_db_results['eval_weighted f1']],
    "RoBERTa (Synonym)": [wordnet_rb_results['eval_f1'], wordnet_rb_results['eval_accuracy'], wordnet_rb_results['eval_weighted f1']],
    "DistilBERT (Back Translation)": [bt_db_results['eval_f1'], bt_db_results['eval_accuracy'], bt_db_results['eval_weighted f1']],
    "RoBERTa (Back Translation)": [bt_rb_results['eval_f1'], bt_rb_results['eval_accuracy'], bt_rb_results['eval_weighted f1']],
    "DistilBERT (Synthetic)": [ft_db_results['eval_f1'], ft_db_results['eval_accuracy'], ft_db_results['eval_weighted f1']],
    "RoBERTa (Synthetic)": [ft_rb_results['eval_f1'], ft_rb_results['eval_accuracy'], ft_rb_results['eval_weighted f1']]
}

comparison_df = pd.DataFrame(data, index=['Macro F1', 'Accuracy', 'Weighted F1'])

display(comparison_df)

# Plotting
comparison_df.T.plot(kind='bar', figsize=(10, 6))
plt.title("Model Performance Comparison")
plt.ylabel("Score")
plt.xticks(rotation=45)
plt.legend(loc='lower right')
plt.show()
```

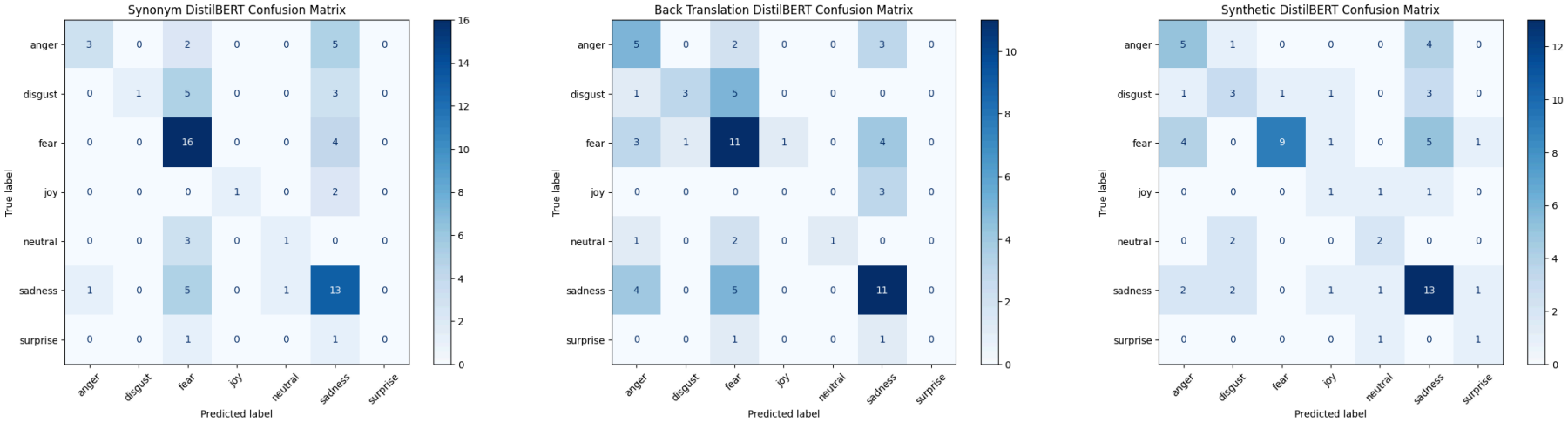
[9/9 00:00]
[9/9 00:00]
[9/9 00:00]
[9/9 00:00]
[9/9 00:09]
[9/9 00:09]

	DistilBERT (Synonym)	RoBERTa (Synonym)	DistilBERT (Back Translation)	RoBERTa (Back Translation)	DistilBERT (Synthetic)	RoBERTa (Synthetic)
Macro F1	0.374137	0.572568	0.325754	0.553487	0.443266	0.484649
Accuracy	0.514706	0.647059	0.455882	0.632353	0.500000	0.573529
Weighted F1	0.471472	0.644235	0.440616	0.627894	0.506782	0.594376



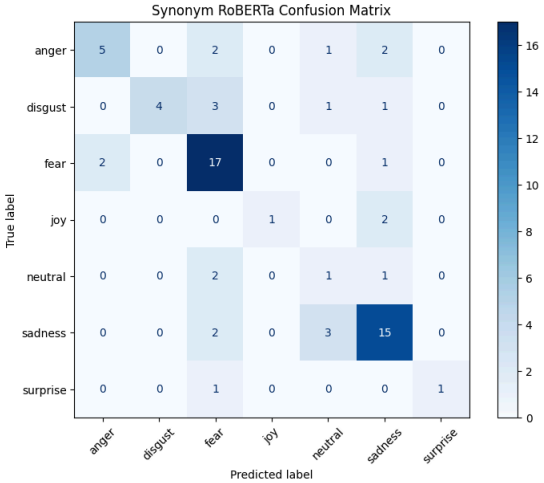
In [68]:
```python
import matplotlib.pyplot as plt
from sklearn.metrics import confusion_matrix, ConfusionMatrixDisplay

def get_predictions(trainer, dataset):
    predictions = trainer.predict(dataset)
    return np.argmax(predictions.predictions, axis=-1)

# Get predictions

wordnet_db_preds = get_predictions(wordnet_trainer_db, val_db)
bt_db_preds = get_predictions(bt_trainer_db, val_db)
ft_db_preds = get_predictions(trainer_db, val_db)

# Get true labels from validation set
y_true = val_df['label_num'].values

# Plotting
fig, (ax1, ax2, ax3) = plt.subplots(1, 3, figsize=(24, 6))

# DistilBERT wordnet
wn_db = confusion_matrix(y_true, wordnet_db_preds)
disp_wn = ConfusionMatrixDisplay(confusion_matrix=wn_db, display_labels=label_list)
disp_wn.plot(ax=ax1, cmap='Blues', xticks_rotation=45)
```

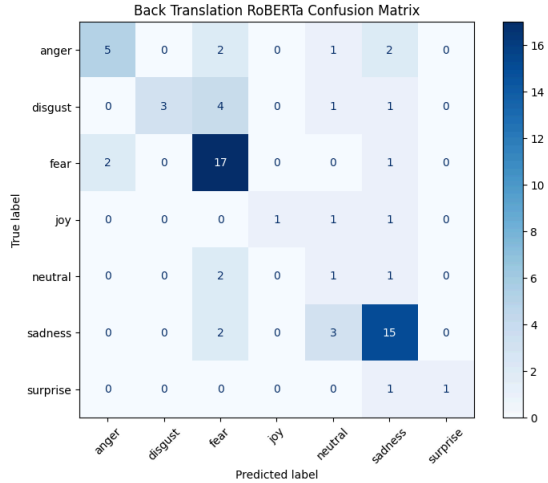
```
ax1.set_title("Synonym DistilBERT Confusion Matrix")

# DistilBERT back translation
bt_db = confusion_matrix(y_true, bt_db_preds)
disp_bt = ConfusionMatrixDisplay(confusion_matrix=bt_db, display_labels=label_list)
disp_bt.plot(ax=ax2, cmap='Blues', xticks_rotation=45)
ax2.set_title("Back Translation DistilBERT Confusion Matrix")

# DistilBERT synthetic
cm_db = confusion_matrix(y_true, ft_db_preds)
disp_db = ConfusionMatrixDisplay(confusion_matrix=cm_db, display_labels=label_list)
disp_db.plot(ax=ax3, cmap='Blues', xticks_rotation=45)
ax3.set_title("Synthetic DistilBERT Confusion Matrix")

# # RoBERTa CM
# cm_rb = confusion_matrix(y_true, ft_rb_preds)
# disp_rb = ConfusionMatrixDisplay(confusion_matrix=cm_rb, display_labels=label_list)
# disp_rb.plot(ax=ax2, cmap='Greens', xticks_rotation=45)
# ax2.set_title("Fine-tuned RoBERTa Confusion Matrix")

plt.tight_layout()
plt.show()
```

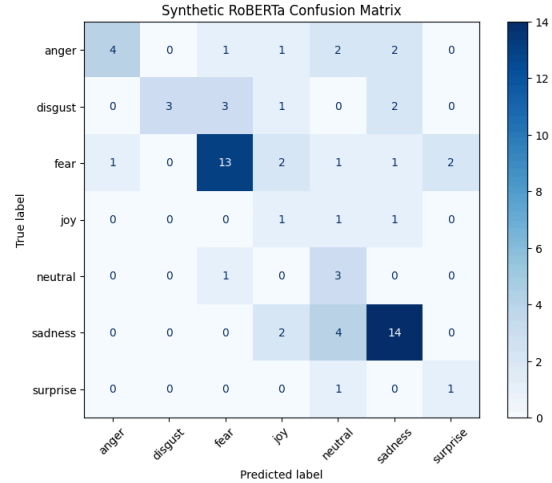


In [69]:

```
import matplotlib.pyplot as plt
from sklearn.metrics import confusion_matrix, ConfusionMatrixDisplay

def get_predictions(trainer, dataset):
    predictions = trainer.predict(dataset)
    return np.argmax(predictions.predictions, axis=-1)

# Get predictions

wordnet_rb_preds = get_predictions(wordnet_trainer_rb, val_rb)
bt_rb_preds = get_predictions(bt_trainer_rb, val_rb)
ft_rb_preds = get_predictions(trainer_rb, val_rb)

# Get true labels from validation set
y_true = val_df['label_num'].values

# Plotting
fig, (ax1, ax2, ax3) = plt.subplots(1, 3, figsize=(24, 6))

# DistilBERT wordnet
wn_rb = confusion_matrix(y_true, wordnet_rb_preds)
disp_wn = ConfusionMatrixDisplay(confusion_matrix=wn_rb, display_labels=label_list)
disp_wn.plot(ax=ax1, cmap='Blues', xticks_rotation=45)
ax1.set_title("Synonym RoBERTa Confusion Matrix")

# DistilBERT back translation
bt_rb = confusion_matrix(y_true, bt_rb_preds)
disp_bt = ConfusionMatrixDisplay(confusion_matrix=bt_rb, display_labels=label_list)
disp_bt.plot(ax=ax2, cmap='Blues', xticks_rotation=45)
ax2.set_title("Back Translation RoBERTa Confusion Matrix")

# DistilBERT synthetic
cm_rb = confusion_matrix(y_true, ft_rb_preds)
disp_db = ConfusionMatrixDisplay(confusion_matrix=cm_rb, display_labels=label_list)
disp_db.plot(ax=ax3, cmap='Blues', xticks_rotation=45)
ax3.set_title("Synthetic RoBERTa Confusion Matrix")

plt.tight_layout()
plt.show()
```


Synonym RoBERTa Confusion Matrix
True label
Predicted label
anger
disgust
fear
joy
neutral
sadness
surprise
5 0 2 0 1 2 0
0 4 3 0 1 1 0
2 0 17 0 0 1 0
0 0 0 1 0 2 0
0 0 2 0 1 1 0
0 0 2 0 3 15 0
0 0 1 0 0 0 1
0
2
4
6
8
10
12
14
16


Back Translation RoBERTa Confusion Matrix
True label
Predicted label
anger
disgust
fear
joy
neutral
sadness
surprise
5 0 2 0 1 2 0
0 3 4 0 1 1 0
2 0 17 0 0 1 0
0 0 0 1 1 1 0
0 0 2 0 1 1 0
0 0 2 0 3 15 0
0 0 0 0 0 1 1
0
2
4
6
8
10
12
14
16


Synthetic RoBERTa Confusion Matrix
True label
Predicted label
anger
disgust
fear
joy
neutral
sadness
surprise
4 0 1 1 2 2 0
0 3 3 1 0 2 0
1 0 13 2 1 1 2
0 0 0 1 1 1 0
0 0 1 0 3 0 0
0 0 0 2 4 14 0
0 0 0 0 1 0 1
0
2
4
6
8
10
12
14